27 April 2026

SQL Aggregate Functions & Operators.

1. List all distinct departments in the students table.

```
SELECT DISTINCT department
FROM students_table;
```

↓

department
IT
HR
Finance

2. Average age of students per department.

```
SELECT department, AVG (age) AS avg_age,
FROM students,
GROUP BY department;
```

↓

department	avg_age
IT	20.5
HR	22.0
Finance	23.0

3. Show departments with more than 1 student.

```
SELECT department, COUNT (student_id) AS student_count
FROM students
GROUP BY department
HAVING COUNT (student_id) > 1;
```

↓

department	student_count
IT	2
HR	2

4. Get all students whose age is between 21 and 23.

```
SELECT student_id,
       name,
       age,
       department
FROM students
WHERE age BETWEEN 21 AND 23;
```

student_id	name	age	department
2	Bob	22	HR
3	charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. List all students in the IT or HR department who are older than 21.

```
SELECT student_id,
       name,
       age,
       department
FROM students
WHERE (department = 'it' OR department = 'hr')
AND age > 21;
```

student_id	name	age	department
2	Bob	22	HR
5	EVE	22	HR

6. Show total credits per department with more than 5 total credits.

```
SELECT department, SUM (credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM (credits) >5 ;
```

↓

department	total_credits
IT	11

7. List all courses that do not have 4 credits.

```
SELECT course_id,
       course_name,
       department,
       credits
  FROM courses
  WHERE credits <> 4;
```

↓

Course_id	course_name	department	credits.
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. Show the top 3 courses by credits in descending order.

```
SELECT course_id,
       course_name,
       credits
  FROM Courses
  ORDER BY credits DESC
  LIMIT 3;
```

↓

Course_id	course_name	credits
102	Python	4
103	Data science	4
101	SQL Basics	3

9. Get the maximum, minimum and average grade across all enrollments.

```
SELECT MAX (grade) AS max_grade,
       MIN (grade) AS min_grade,
       AVG (grade) AS avg_grade
FROM enrollments;
```

↓

max_grade	min_grade	avg_grade
90	78	84.6

10. Count how many enrollments exist per course.

```
SELECT course_id, COUNT (enrollment_id) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

↓

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11. Find total salary and total bonus per department.

```
SELECT department, SUM (salary) AS total_salary,
       SUM (bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

↓

department	total_salary	total_bonus
IT	122 000	10500
HR	109 000	7500
Finance	70 000	6000

12. Show departments where average salary is above 55,000.

```
SELECT department, AVG (salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG (salary) > 55 000;
```

↓

department	avg_salary
IT	61 000
Finance	70 000

13. List employees where salary plus bonus is greater than 60,000.

```
SELECT employee_id,
       name,
       salary,
       bonus,
       (salary + bonus) AS total_compensation
FROM salaries
WHERE (salary + bonus) > 60 000;
```

↓

employee_id	name	salary	bonus	total compensation.
1	Tom	60 000	5000	65 000
3	Spike	70 000	6000	76 000
4	Tyke	62 000	5500	67 500

14. show total and average budget per department. Only include departments with average budget above 70 000.

```
SELECT department SUM (budget) AS total_budget,
       AVG (budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70000;
```

↓

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. List all projects with budgets between 50,000 and 120,000 excluding the marketing department.

```
SELECT project_id,
       project_name,
       department,
       budget
FROM projects
WHERE (budget BETWEEN 50000 AND 120 000)
AND department <> 'Marketing';
```

↓

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50000